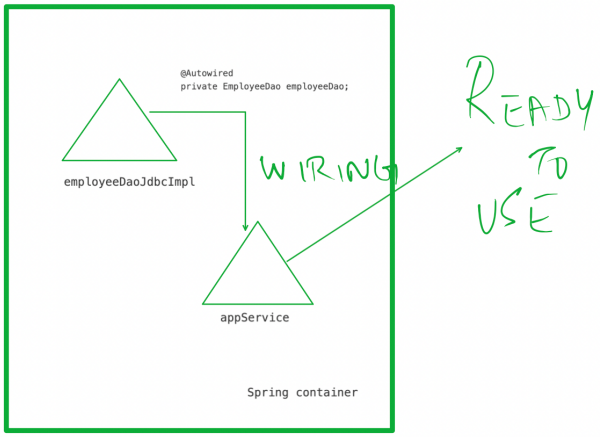
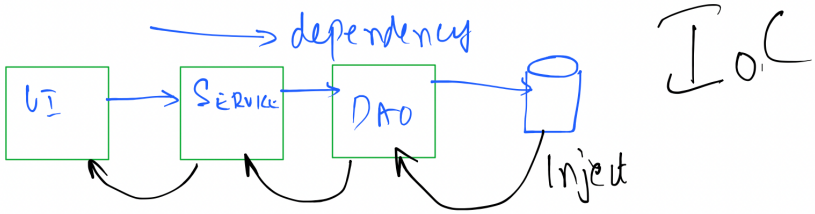
```java
public interface EmployeeDao {
    void addEmployee(Employee e);
}

@Repository
public class EmployeeDaoJdbcImpl implements EmployeeDao {
    ...
}

@Service
public class AppService {
    @Autowired
    private EmployeeDao employeeDao;

    public void task(Employee e) {
        employeeDao.addEmployee(e);
    }
}
```



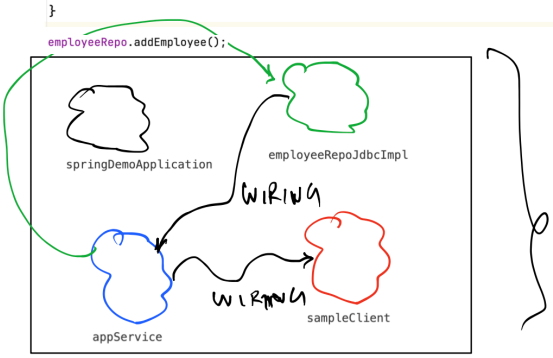


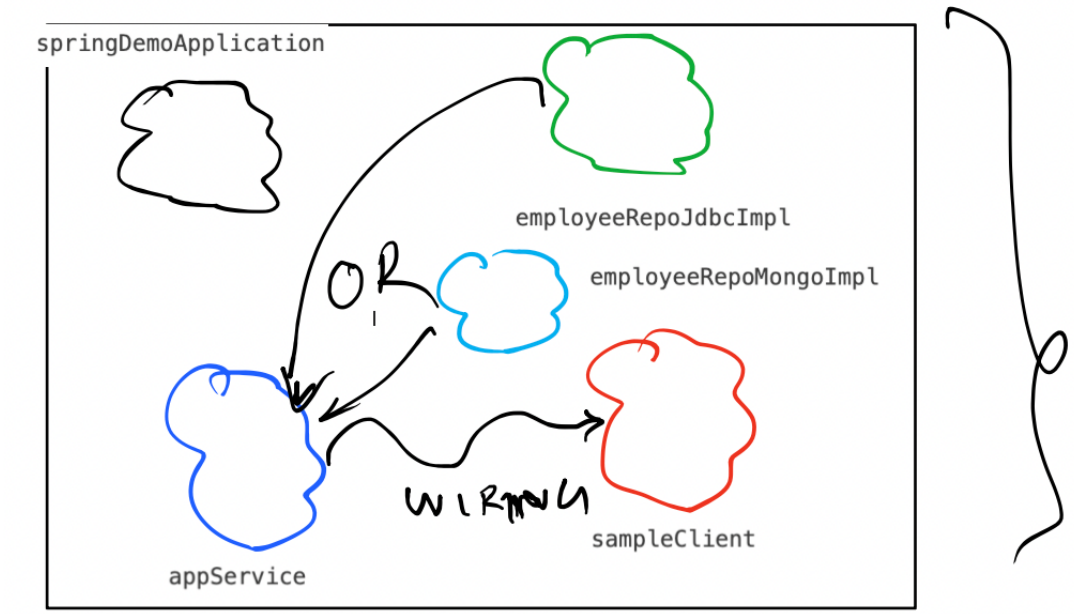

```java
@Repository
public class EmployeeRepoJdbcImpl implements EmployeeRepo{
    @Override
    public void addEmployee() {
        System.out.println("Stored in database!!!");
    }
}
```

```java
@Component
public class SampleClient implements CommandLineRunner {
    @Autowired
    private AppService service;
    // executes as soon as spring container is created and initialized
    @Override
    public void run(String... args) throws Exception {
        service.doTask();
    }
}
```

```java
@Service
public class AppService {
    @Autowired
    private EmployeeRepo employeeRepo;

    public void doTask() {
        employeeRepo.addEmployee();
    }
}
```

```java
@SpringBootApplication
public class SpringDemoApplication {
    public static void main(String[] args) {
        // this line is what creates Spring Container
        SpringApplication.run(SpringDemoApplication.class, args);
    }
}
```






```
Field employeeRepo in AppService required a single bean, but 2 were found:
    - employeeRepoJdbcImpl:
    - employeeRepoMongoImpl:
```

Solution:

```java
@Service
public class AppService {
    @Autowired
    @Qualifier("employeeRepoJdbcImpl")
    private EmployeeRepo employeeRepo;
```

JDBC

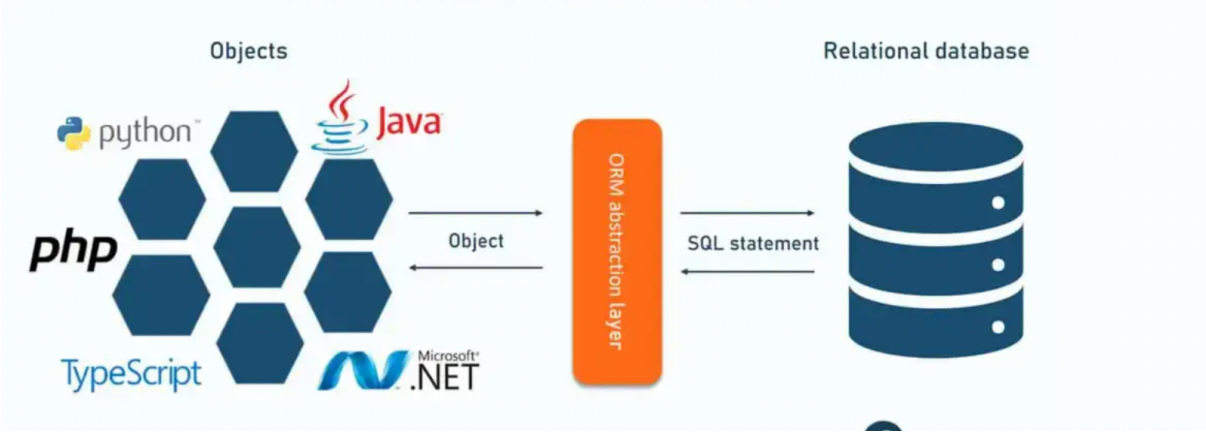
```java
@Override
public void addProduct(Product product) throws PersitenceException {
    Connection con = null;
    String SQL = "INSERT INTO products(id, name, price) VALUES (0, ?, ?)";
    try {
        con = DriverManager.getConnection(URL, USER, PWD);
        PreparedStatement statement = con.prepareStatement(SQL);
        statement.setString(1, product.getName());
        statement.setDouble(2, product.getPrice());
        statement.executeUpdate(); // INSERT, DELETE or UPDATE
    } catch (SQLException ex) {
        if (ex.getErrorCode() == 1062) {
            throw new PersitenceException("unable to add product " + product.getId() + " already exists!!", ex);
        } else if (ex.getErrorCode() == 1054) {
            throw new PersitenceException("unable to add product, Bad Input !!!", ex);
        }
        else{
        throw new PersitenceException("unable to add product " + product + " !!!", ex);
    }
}
    finally {
        if( con != null) {
            try {
                con.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

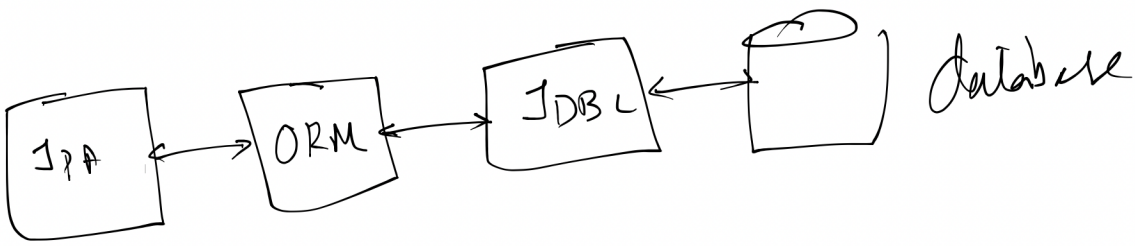
ORM

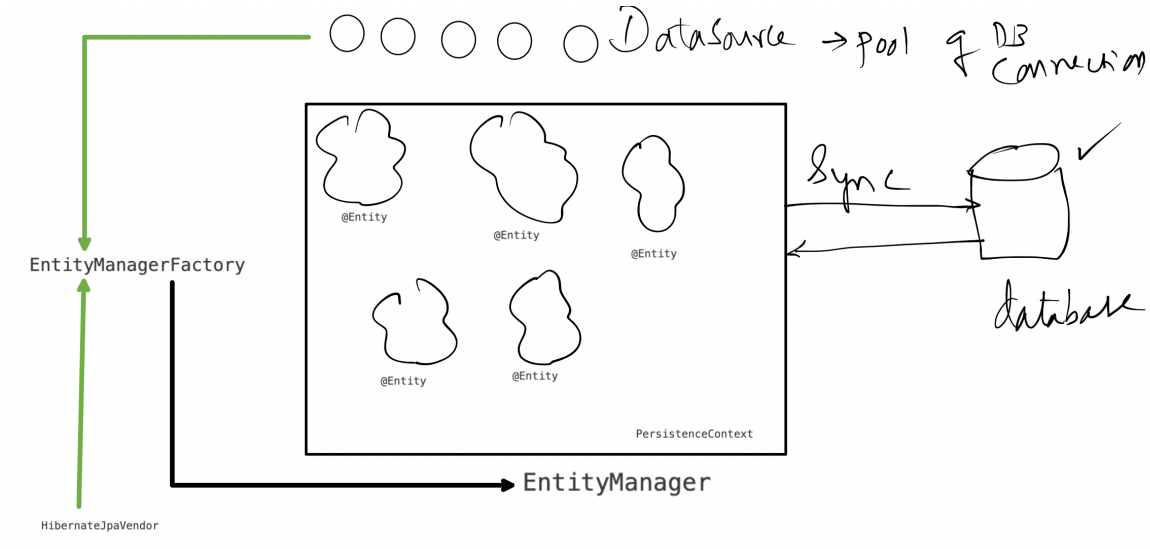
```java
@Override
public void addProduct(Product product)  {
    em.persist(product);
}
```

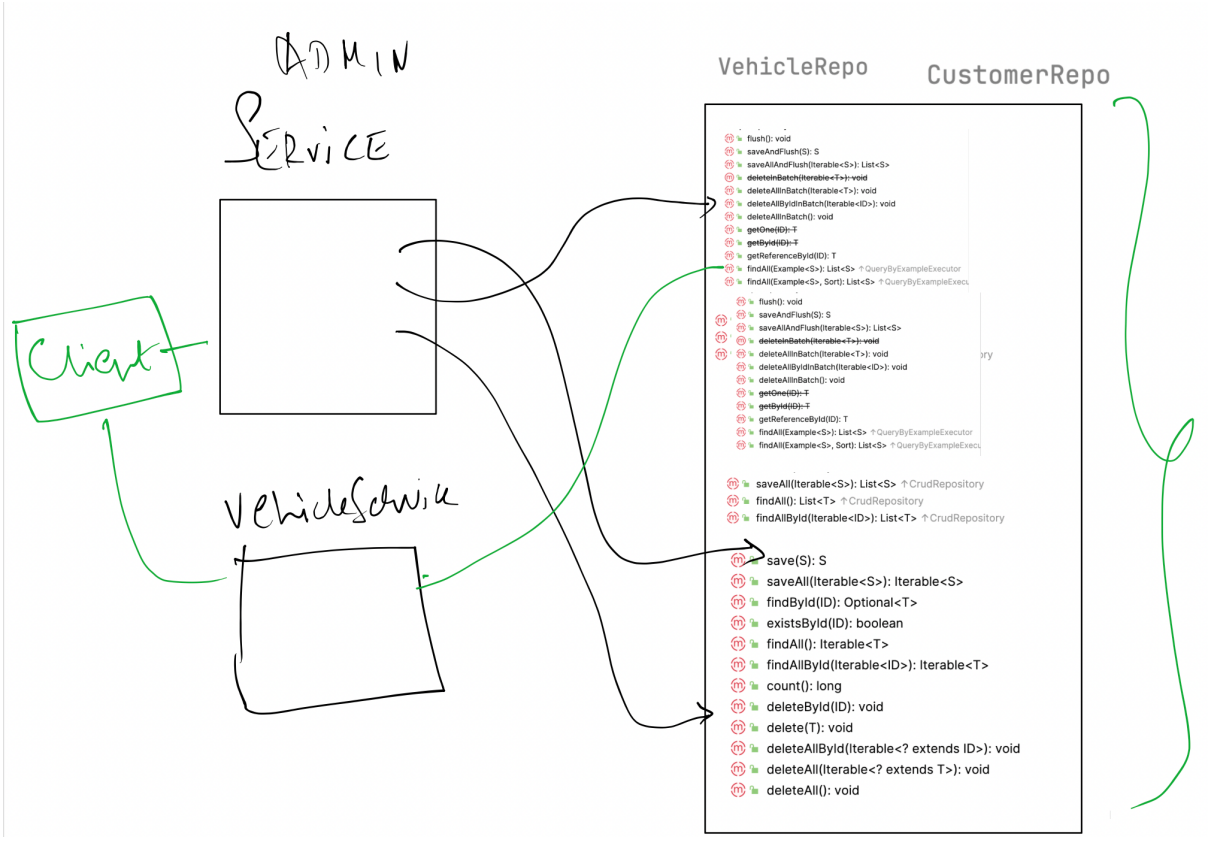
```java
@Entity
@Table(name="products")
public class Product {
    @Id
    private int id;

    private String name;

    private double price;
}
```

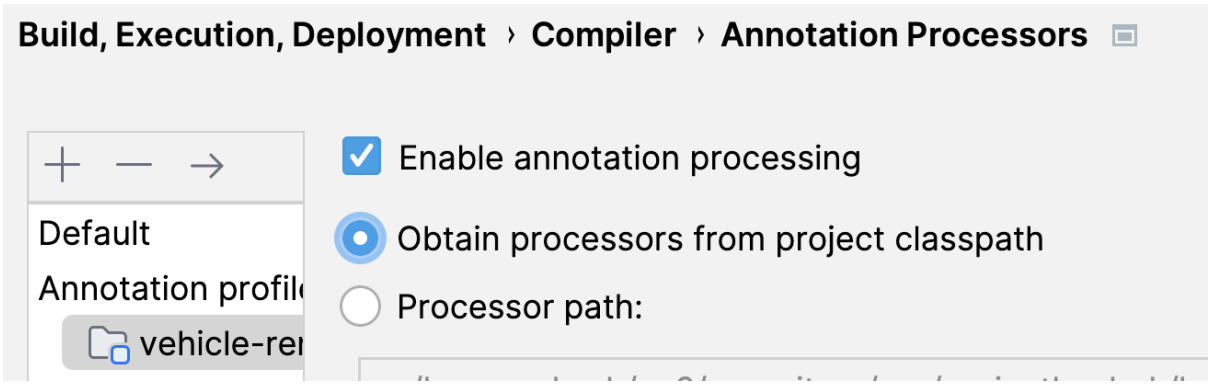
https://docs.oracle.com/javase/8/docs/api/java/sql/Types.html








ADMIN
SERVICE
Client
VehicleService
VehicleRepo
CustomerRepo
flush(): void
saveAndFlush(S): S
saveAllAndFlush(Iterable<S>): List<S>
deleteInBatch(Iterable<T>): void
deleteAllInBatch(Iterable<T>): void
deleteAllByIdInBatch(Iterable<ID>): void
deleteAllInBatch(): void
getOne(ID): T
getById(ID): T
getReferenceById(ID): T
findAll(Example<S>): List<S> ↑QueryByExampleExecutor
findAll(Example<S>, Sort): List<S> ↑QueryByExampleExecu
saveAll(Iterable<S>): List<S> ↑CrudRepository
findAll(): List<T> ↑CrudRepository
findAllById(Iterable<ID>): List<T> ↑CrudRepository
save(S): S
saveAll(Iterable<S>): Iterable<S>
findById(ID): Optional<T>
existsById(ID): boolean
findAll(): Iterable<T>
findAllById(Iterable<ID>): Iterable<T>
count(): long
deleteById(ID): void
delete(T): void
deleteAllById(Iterable<? extends ID>): void
deleteAll(Iterable<? extends T>): void
deleteAll(): void

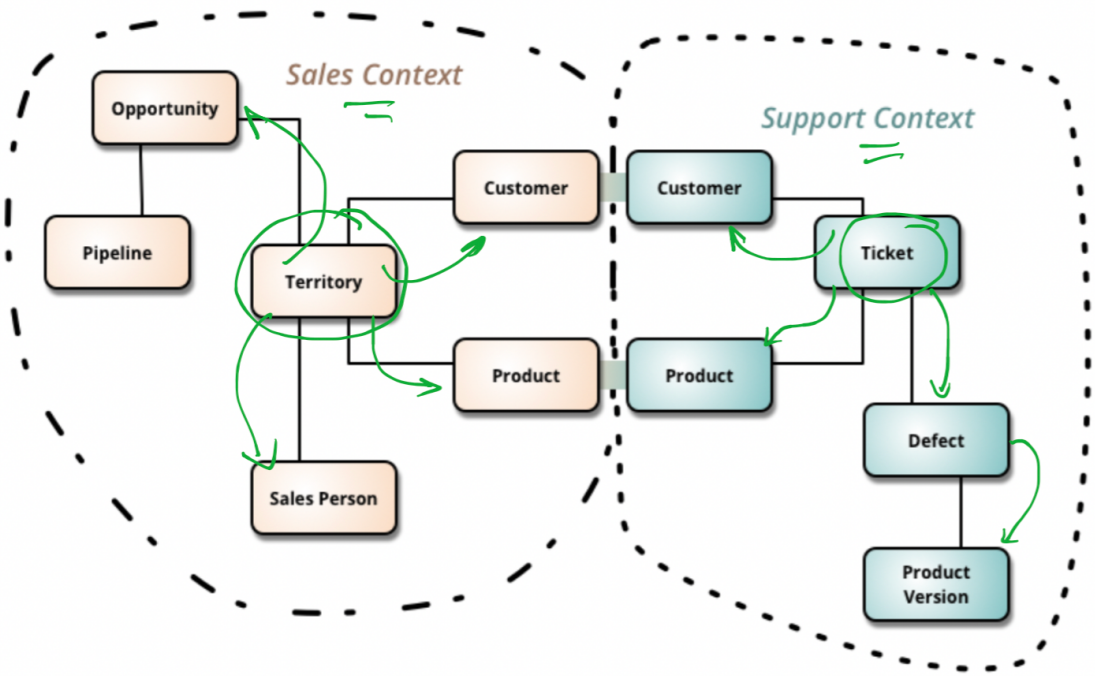

Build, Execution, Deployment › Compiler › Annotation Processors
Default
Annotation profil
vehicle-re
Enable annotation processing
Obtain processors from project classpath
Processor path:

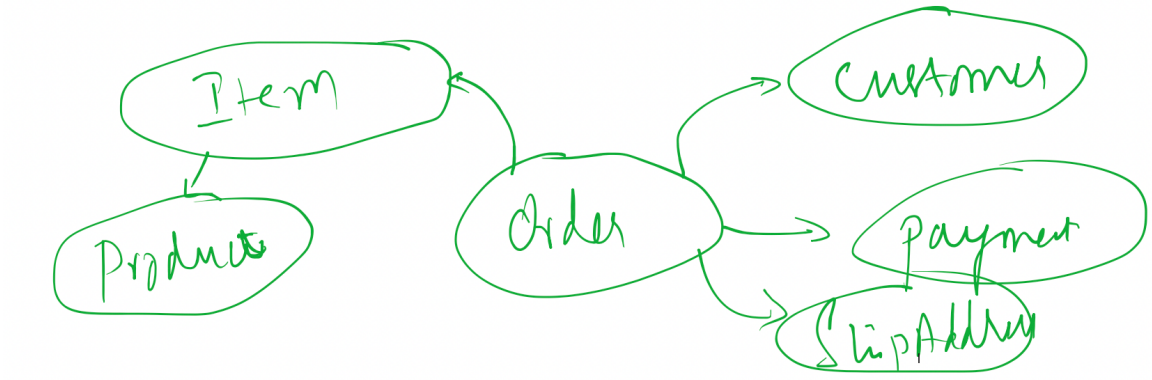
```
+------------------+--------+----------+
| email            | fname  | lname    |
+------------------+--------+----------+
| anne@adobe.com   | Anne   | Hathaway |
| roger@adobe.com  | Roger  | Smith    |
+------------------+--------+----------+
```

```
+---------------+-----------+-----------+
| reg_no        | hire_rate | fuel_type |
+---------------+-----------+-----------+
| DH-10-AA-0434 |      5300 | PETROL    |
| KA-05-AB-1234 |      4500 | PETROL    |
| UP-15-EB-4321 |      6500 | ELECTRIC  |
+---------------+-----------+-----------+
```

bookings

```
id | customer_fk       | vehicle_fk     | date_from.  | date_to    | amount

23  | anne@adobe.com.  | UP-15-EB-4321 |  18-AUG-2025. | null.        | 0
24  | roger@adobe.com. | KA-05-AB-1234 |  18-AUG-2025. | 20-AUG-2025 | 8900.00
25 | roger@adobe.com |  DH-10-AA-0434 | 21-AUG-2025 | null | 0
26 | peter@adobe.com | KA-05-AB-1234 | 21-AUG-2025 | null | 0
```






```
mysql> select * from customers;
+------------------+-------+-------+
| email            | fname | lname |
+------------------+-------+-------+
| danny@adobe.com  | Danny | Smith |
| rita@adobe.com   | Rita  | Smith |
+------------------+-------+-------+
```

```
mysql> select * from products;
+----+-------------+--------+----------+
| id | name        | price  | quantity |
+----+-------------+--------+----------+
|  1 | iPhone 16   |  89000 |       98 |
|  2 | Sony Bravia | 289000 |       99 |
|  3 | LG AC       |  54000 |      100 |
+----+-------------+--------+----------+
```

```
mysql> select * from orders;
+-----+----------------------------+--------+------------------+
| oid | order_date                 | total  | customer_fk      |
+-----+----------------------------+--------+------------------+
|   1 | 2025-08-21 16:58:05.746000 | 467000 | rita@adobe.com |
+-----+----------------------------+--------+------------------+
```

```
mysql> select * from items;
+---------+--------+-----+------------+----------+
| item_id | amount | qty | product_fk | order_fk |
+---------+--------+-----+------------+----------+
|       1 | 289000 |   1 |          2 |        1 |
|       2 | 178000 |   2 |          1 |        1 |
+---------+--------+-----+------------+----------+
```